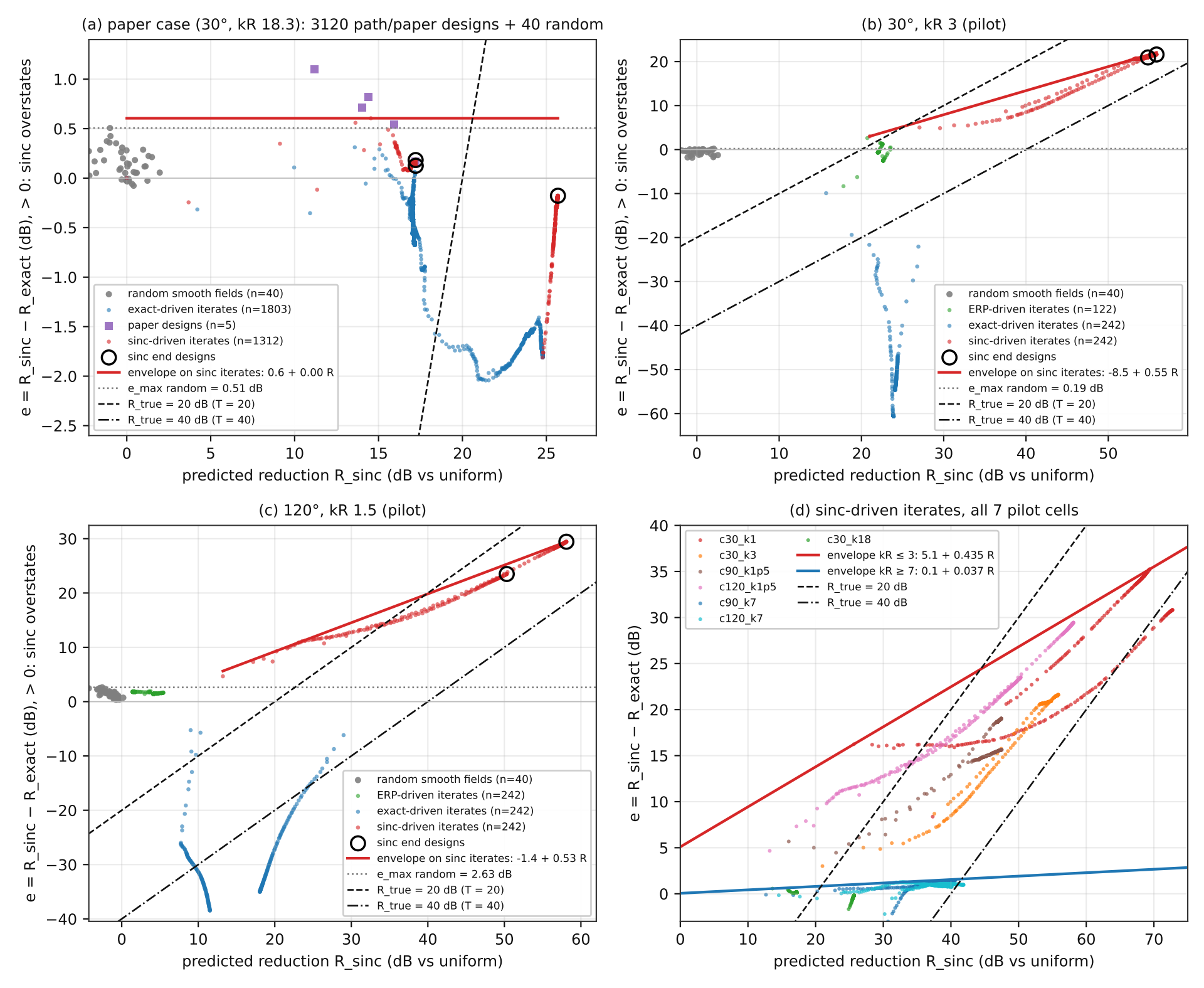

(a) paper case (30°, kR 18.3): 3120 path/paper designs + 40 random
e = R_sinc − R_exact (dB), > 0: sinc overstates
predicted reduction R_sinc (dB vs uniform)
random smooth fields (n=40)
exact-driven iterates (n=1803)
paper designs (n=5)
sinc-driven iterates (n=1312)
sinc end designs
envelope on sinc iterates: 0.6 + 0.00 R
e_max random = 0.51 dB
R_true = 20 dB (T = 20)
R_true = 40 dB (T = 40)
(b) 30°, kR 3 (pilot)
e = R_sinc − R_exact (dB), > 0: sinc overstates
predicted reduction R_sinc (dB vs uniform)
random smooth fields (n=40)
ERP-driven iterates (n=122)
exact-driven iterates (n=242)
sinc-driven iterates (n=242)
sinc end designs
envelope on sinc iterates: -8.5 + 0.55 R
e_max random = 0.19 dB
R_true = 20 dB (T = 20)
R_true = 40 dB (T = 40)
(c) 120°, kR 1.5 (pilot)
e = R_sinc − R_exact (dB), > 0: sinc overstates
predicted reduction R_sinc (dB vs uniform)
random smooth fields (n=40)
ERP-driven iterates (n=242)
exact-driven iterates (n=242)
sinc-driven iterates (n=242)
sinc end designs
envelope on sinc iterates: -1.4 + 0.53 R
e_max random = 2.63 dB
R_true = 20 dB (T = 20)
R_true = 40 dB (T = 40)
(d) sinc-driven iterates, all 7 pilot cells
e = R_sinc − R_exact (dB)
predicted reduction R_sinc (dB vs uniform)
c30_k1
c30_k3
c90_k1p5
c120_k1p5
c90_k7
c120_k7
c30_k18
envelope kR ≤ 3: 5.1 + 0.435 R
envelope kR ≥ 7: 0.1 + 0.037 R
R_true = 20 dB
R_true = 40 dB